Anil Nair, CTO (left), Llewellyn D'sa, CEO, and Priyanka D'sa, COO

Person fitted with Grippy reading a newspaper

Grippy on a weighing scale

Indian market were either very expensive or were of low quality. Moreover, the repair of products was either not possible or was very expensive.

So, in order to help people with a missing arm, Llewellyn decided to take up research and development of indigenous bionic arms as his M. Tech project. He and his friends, who helped him with the project, started Robo Bionics. From the first prototype to the final product, the journey took them around three and a half years and almost ten iterations to reach where they are today.

Robo Bionics tested their bionic arms on ten disabled people who were chosen for the trials. Since then, the company has fitted bionic arms to more than 50 people across India.

The current bionic arm from Robo Bionics is called Grippy. It is a lightweight, affordable, battery-powered prosthetic arm, which is suitable for anyone over the age of 15 years with below-elbow amputation.

Unlike most bionic arms available in the market, which can't cater to people with different muscle strengths, Grippy can be programmed based on the user's muscle strength. This gives them accurate control over the device.

Grippy can enable the user to sense objects. Moreover, it can differentiate between a soft and a hard object and offers an automatically adjusted grip control, which enables the user to hold an object without damaging it with excessive force. The ability to program Grippy makes it very easy to use; most people are able to control it on the very first day.

Imported bionics arms made of titanium weigh around 600 grams and cost around a million rupees. Grippy having a similar weight costs one-fourth of that, making smart prosthetics arms accessible to a larger population in India.

Robo Bionics employs 3D printing technology to print the entire shell of the bionic arm. They use a proprietary thermoplastic material, which is lightweight but has strength comparable to the human arms. Grippy can sustain temperatures up to 125°C and can lift up to 5kg in a hook grip, which is required for holding a bag or a briefcase.

Llewellyn D'sa says, "Currently, the hand fitting process takes two to three weeks. With the upgradation of technology and utilisation of additive manufacturing, we could actually see the process complete in a matter of may be two to three days."

Being a bionic arm, Grippy employs various sensors and actuators to sense and hold objects. It contains a microcontroller peripheral unit, current sensing ICs, the inertial measurement unit (IMU), etc. Besides, there is an indicator system, operational amplifiers, isolation amplifiers, and the display system. As each electronic subsystem of the arm needs its own power source, power regulators and batteries to power the whole system are essential for it.

Llewellyn believes that, in future, the bionic arm will become lighter and will have better battery efficiency, better power delivery, and IoT connectivity. Integration of IoT will give proper feedback to the patient as well as the caregiver or a doctor. Apart from that there will be an integration of artificial intelligence, which will help the users to have proper coordination and application of force based on the object.

He hopes that insurance companies would step up their game and include rehabilitation technology as a point of care system. He also looks up to the government to launch new schemes to help people with disabilities. With such support, the company hopes to succeed in empowering individuals suffering from limb loss to regain their independence and achieve their full potential. EFY

—Sharad Bhowmick